

积分方程

1. 设函数 $f(x)$ 在区间 $[0,1]$ 上连续, 且 $f(x) = x^2 + 2\sqrt{x} \int_0^1 f(x)dx$, 求 $f(x)$.

解 (注意 $\int_0^1 f(x)dx$ 是常数, 设常数去积分号)

令 $A = \int_0^1 f(x)dx$, 则 $f(x) = x^2 + 2A\sqrt{x}$, 于是

$$A = \int_0^1 f(x)dx = \int_0^1 (x^2 + 2A\sqrt{x})dx = \left(\frac{x^3}{3} + \frac{4A}{3}x^{\frac{3}{2}} \right) \Big|_0^1 = \frac{1}{3} + \frac{4A}{3}$$

解得 $A = -1$, 故

$$f(x) = x^2 - 2\sqrt{x}$$

2. 连续函数 $f(x)$ 满足 $f(x) - \cos^2 x = \frac{1}{\pi} \int_0^{\frac{\pi}{2}} f(2t)dt$, 计算 $\int_0^{\frac{\pi}{2}} f(x)dx$.

解 令 $u = 2t$, 则 $t = \frac{u}{2}$, $dt = \frac{1}{2}du$, 当 $t = 0$ 时 $u = 0$ (积分下限), 当 $t = \frac{\pi}{4}$ 时 $u = \frac{\pi}{2}$

(积分上限), 所以

$$\int_0^{\frac{\pi}{4}} f(2t)dt = \int_0^{\frac{\pi}{2}} f(u) \left(\frac{1}{2}du \right) = \frac{1}{2} \int_0^{\frac{\pi}{2}} f(u)du = \frac{1}{2} \int_0^{\frac{\pi}{2}} f(x)dx$$

代入方程得

$$f(x) - \cos^2 x = \frac{1}{2\pi} \int_0^{\frac{\pi}{2}} f(x)dx$$

(注意 $\int_0^{\frac{\pi}{2}} f(x)dx$ 是常数, 设常数去积分号)

令 $A = \int_0^{\frac{\pi}{2}} f(x)dx$, 则 $f(x) = \cos^2 x + \frac{A}{2\pi}$, 于是

$$\begin{aligned} A &= \int_0^{\frac{\pi}{2}} f(x)dx = \int_0^{\frac{\pi}{2}} \left(\cos^2 x + \frac{A}{2\pi} \right) dx = \int_0^{\frac{\pi}{2}} \cos^2 x dx + \int_0^{\frac{\pi}{2}} \frac{A}{2\pi} dx \\ &= \frac{1}{2} \cdot \frac{\pi}{2} + \frac{A}{2\pi} \cdot \frac{\pi}{2} = \frac{\pi}{4} + \frac{A}{4} \end{aligned}$$

解得 $A = \frac{\pi}{3}$, 所以

$$\int_0^{\frac{\pi}{2}} f(x)dx = A = \frac{\pi}{3} \quad \left(\text{此时 } f(x) = \cos^2 x + \frac{1}{6} \right)$$

3. 设函数 $f(x)$ 连续, 且满足 $\int_0^x e^{f(x)} dx = xe^{2x} - x^2 f'(0)$, 求 $f(x)$.

解 ($\int_0^x e^{f(x)} dx = \int_0^x e^{f(t)} dt$ 是变量即函数, 求导去积分号)

方程两边求导

$$\left(\int_0^x e^{f(x)} dx \right)' = (xe^{2x} - x^2 f'(0))'$$

得

$$e^{f(x)} = e^{2x} + x \cdot 2e^{2x} - 2xf'(0) \Rightarrow e^{f(x)} = (2x+1)e^{2x} - 2xf'(0)$$

取对数得

$$f(x) = \ln[(2x+1)e^{2x} - 2xf'(0)]$$

为求 $f'(0)$, 求导得

$$f'(x) = \frac{1}{(2x+1)e^{2x} - 2xf'(0)} [2e^{2x} + (2x+1) \cdot 2e^{2x} - 2f'(0)]$$

令 $x=0$ 得 $f'(0) = \frac{1}{1} \cdot [4 - 2f'(0)]$, 解得 $f'(0) = \frac{4}{3}$, 故

$$f(x) = \ln \left[(2x+1)e^{2x} - \frac{8}{3}x \right]$$

4. 设函数 $f(x)$ 及其反函数 $\varphi(x)$ 都可导, 且满足关系式 $\int_1^{f(x)} \varphi(t) dt = \frac{1}{3} \left(x^{\frac{3}{2}} - 8 \right)$, 求

$f(x)$.

解 ($\int_1^{f(x)} \varphi(t) dt$ 是变量即函数, 求导去积分号)

方程两边求导

$$\left(\int_1^{f(x)} \varphi(t) dt \right)' = \left[\frac{1}{3} \left(x^{\frac{3}{2}} - 8 \right) \right]'$$

得

$$\varphi[f(x)]f'(x) = \frac{1}{2}\sqrt{x}$$

因为 $\varphi(x)$ 是 $f(x)$ 的反函数, 所以 $\varphi[f(x)] = x$, 于是上式化为

$$xf'(x) = \frac{1}{2}\sqrt{x} \Rightarrow f'(x) = \frac{1}{2\sqrt{x}} \Rightarrow f(x) = \sqrt{x} + C$$

在所给方程中令 $f(x)=1$ 得 $0=\frac{1}{3}\left(x^{\frac{3}{2}}-8\right)$, 解得 $x=4$, 所以 $f(4)=1$, 由此得

$$1=\sqrt{4}+C \Rightarrow C=-1, \text{ 从而}$$

$$f(x)=\sqrt{x}-1$$

5. 设函数 $f(x)$ 连续, 且 $\int_0^1 f(xt)dt = \frac{x^2}{3} - \frac{x}{2} \int_0^2 f(x)dx + 2 \int_0^1 f(x)dx$, 求 $f(x)$.

解 (注意到 $\int_0^1 f(xt)dt$ 是变量即函数, $\int_0^1 f(x)dx$ 和 $\int_0^2 f(x)dx$ 是常数)

当 $x \neq 0$ 时, 令 $u=xt$, 则 $t=\frac{u}{x}, dt=\frac{1}{x}du$, 当 $t=0$ 时 $u=0$ (积分下限), 当 $t=1$ 时

$u=x$ (积分上限), 所以

$$\int_0^1 f(xt)dt = \int_0^x f(u)\left(\frac{1}{x}du\right) = \frac{1}{x} \int_0^x f(u)du$$

代入方程得

$$\frac{1}{x} \int_0^x f(u)du = \frac{x^2}{3} - \frac{x}{2} \int_0^2 f(x)dx + 2 \int_0^1 f(x)dx$$

由此得

$$\int_0^x f(u)du = \frac{x^3}{3} - \frac{x^2}{2} \int_0^2 f(x)dx + 2x \int_0^1 f(x)dx$$

上式 $x=0$ 时也成立, 求导得

$$f(x) = x^2 - x \int_0^2 f(x)dx + 2 \int_0^1 f(x)dx$$

令 $A = \int_0^2 f(x)dx, B = \int_0^1 f(x)dx$, 则

$$f(x) = x^2 - Ax + 2B$$

于是

$$A = \int_0^2 f(x)dx = \int_0^2 (x^2 - Ax + 2B)dx = \left(\frac{x^3}{3} - \frac{Ax^2}{2} + 2Bx\right)\Big|_0^2 = \frac{8}{3} - 2A + 4B$$

$$B = \int_0^1 f(x)dx = \int_0^1 (x^2 - Ax + 2B)dx = \left(\frac{x^3}{3} - \frac{Ax^2}{2} + 2Bx\right)\Big|_0^1 = \frac{1}{3} - \frac{A}{2} + 2B$$

$$\text{解方程组} \begin{cases} A = \frac{8}{3} - 2A + 4B \\ B = \frac{1}{3} - \frac{A}{2} + 2B \end{cases} \quad \text{得 } A = \frac{4}{3}, B = \frac{1}{3}, \text{ 因此}$$

$$f(x) = x^2 - \frac{4}{3}x + \frac{2}{3}$$

6. 设函数 $f(x)$ 在区间 $(-\infty, +\infty)$ 上连续, 在点 $x=0$ 处可导, 且满足方程

$$f(x) = x^3 + x^2 \lim_{x \rightarrow 0} \frac{f(x)}{x} - x \int_0^1 f(x) dx, \text{ 求函数 } f(x) \text{ 的表达式.}$$

解 易知 $f(0) = 0$, 所以

$$\lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = f'(0)$$

记 $a = f'(0)$, $b = \int_0^1 f(x) dx$, 则

$$f(x) = x^3 + ax^2 - bx$$

于是

$$a = f'(0) = (3x^2 + 2ax - b)|_{x=0} = -b$$

$$b = \int_0^1 f(x) dx = \int_0^1 (x^3 + ax^2 - bx) dx = \frac{1}{4} + \frac{a}{3} - \frac{b}{2}$$

即有

$$\begin{cases} a = -b \\ b = \frac{1}{4} + \frac{a}{3} - \frac{b}{2} \end{cases}$$

解得 $a = -\frac{3}{22}$, $b = \frac{3}{22}$, 所以

$$f(x) = x^3 - \frac{3}{22}x^2 - \frac{3}{22}x$$

7. 设函数 $f(x)$ 连续, 且满足 $\int_0^x t f(2x-t) dt = \frac{1}{2} \arctan x^2$, $f(1) = 1, f(2) = 3$,

计算 $\int_1^2 f(x) dx$ 和 $\int_1^2 x f(x) dx$.

解 令 $u = 2x - t$, 则 $t = 2x - u, dt = -du$, 当 $t = 0$ 时 $u = 2x$ (积分下限), 当 $t = x$ 时 $u = x$ (积分上限), 所以

$$\begin{aligned}\int_0^x t f(2x - t) dt &= \int_{2x}^x (2x - u) f(u) (-du) \\ &= \int_x^{2x} (2x - u) f(u) du = 2x \int_x^{2x} f(u) du - \int_x^{2x} u f(u) du\end{aligned}$$

代入原方程得

$$2x \int_x^{2x} f(u) du - \int_x^{2x} u f(u) du = \frac{1}{2} \arctan x^2$$

为了消去积分 $\int_x^{2x} u f(u) du$, 求导得

$$2 \int_x^{2x} f(u) du + 2x [f(2x) \cdot 2 - f(x)] - [2xf(2x) \cdot 2 - xf(x)] = \frac{x}{1+x^4}$$

解得

$$\int_x^{2x} f(u) du = \frac{1}{2} \left(xf(x) + \frac{x}{1+x^4} \right)$$

令 $x = 1$ 得

$$\int_1^2 f(u) du = \frac{1}{2} \left(f(1) + \frac{1}{2} \right) = \frac{1}{2} \left(1 + \frac{1}{2} \right) = \frac{3}{4}$$

$$\text{即 } \int_1^2 f(x) dx = \frac{3}{4}.$$

为了消去积分 $\int_x^{2x} f(u) du$, 将方程写成

$$\int_x^{2x} f(u) du - \frac{1}{2x} \int_x^{2x} u f(u) du = \frac{1}{4x} \arctan x^2$$

求导得

$$\begin{aligned}& [f(2x) \cdot 2 - f(x)] + \frac{1}{2x^2} \int_x^{2x} u f(u) du - \frac{1}{2x} [2xf(2x) \cdot 2 - xf(x)] \\ &= -\frac{1}{4x^2} \arctan x^2 + \frac{1}{4x} \frac{2x}{1+x^4}\end{aligned}$$

令 $x = 1$ 得

$$[2f(2) - f(1)] + \frac{1}{2} \int_1^2 u f(u) du - \frac{1}{2} [4f(2) - f(1)] = -\frac{\pi}{16} + \frac{1}{4}$$

$$\text{将 } f(1) = 1, f(2) = 3 \text{ 代入得 } \int_1^2 u f(u) du = \frac{3}{2} - \frac{\pi}{8}, \text{ 即 } \int_1^2 x f(x) dx = \frac{3}{2} - \frac{\pi}{8}.$$